

[This question paper contains 5 printed pages.]

1467-A

Your Roll No.

B.A./B.Sc. (Hons.)/II

A

MATHEMATICS – Unit VIII

(Numerical Analysis and Computer Programming)

Time : 2. Hours

Maximum Marks : 30

*(Write your Roll No. on the top immediately
on receipt of this question paper.)*

*All questions are compulsory.
Choice is given within the question.
Use of Scientific Calculator is allowed.*

SECTION I

1. (a) Explain executable and non-executable statements with examples. (1)

- (b) Write the FORTRAN equivalent of

$$\frac{x^3 + y^3}{\cos(x + y)} + e^{xy} - \sqrt{|x^2 - y^2|} \quad (1)$$

- (c) Evaluate the expression

MOD(INT(23 - 4.3), I * I) . LT . K .

OR . I + J/K . GT . J - 11

where I = 2, J = 15, K = 6 (1)

P.T.O.

- (d) The following program calculates the circumference and area of a circle when the radius is given. The program contains many syntax errors. Rewrite the same program after removing the syntax errors.

1	6
C	Program to find the circumference
	+ & area of a circle.
	+ Radius is given.
	Read (*, *) R
	Real, R
	WRITE *, 'Radius is' R
	CIRCUMFE = $2\pi R$
	AR = $\pi R ** 2$
	WRITE *, CIRCUMFE, AREA
	STOP.

(2)

- (e) Write a program in FORTRAN 77 to find the mean and standard deviation of a given set of 20 numbers.

OR

Write a subroutine subprogram in FORTRAN 77 to test whether a given natural number is prime.

(2)

SECTION II.

2. Attempt any two parts :

(a) Using secant method find a positive root of $x^3 + x^2 - 3x - 3 = 0$. Do 3 iterations. ~~Also write a program to implement it.~~ (4)

(b) What do you understand by the term 'rate of convergence' of an iterative method for solving an equation? Derive the rate of convergence of the Newton Raphson Method for solving an equation iteratively. (4)

(c) The equation $x^2 + ax + b = 0$ has two real roots α and β , show that

(i) the iteration method $x_{n+1} = -(ax_n + b)/x_n$ is convergent near $x = \alpha$ if $|\alpha| > |\beta|$

(ii) the iteration method $x_{n+1} = -\frac{b}{x_n + a}$ is convergent near $x = \alpha$ if $|\alpha| < |\beta|$

(iii) the iteration method $x_{n+1} = -\frac{(x_n^2 + b)}{a}$ is convergent near $x = \alpha$ if $2|\alpha| < |\alpha + \beta|$. (4)

SECTION III

3. Attempt any two parts :

(a) Find the inverse of the coefficient matrix of the system of equations

$$x + y + z = 1$$

$$4x + 3y - z = 6$$

$$3x + 5y + 3z = 4$$

by Gauss Jordan method with partial pivoting.
Hence solve the system. (4)

(b) For the system of equations

$$4x - y + z = 7$$

$$4x - 8y + z = -21$$

$$-2x + y + 5z = 15$$

set up the Gauss Jacobi Iterative Scheme. Do 3 iterations, using a suitable initial approximation.

(4)

(c) Write a program in FORTRAN 77 to solve the system of equations

$$22x_1 + x_2 - x_3 - x_4 = 12$$

$$2x_1 + 13x_2 - x_3 + 2x_4 = 21$$

$$2x_2 + 30x_3 - 5x_4 = 11$$

$$x_1 + x_2 + x_3 - 6x_4 = -5$$

by Gauss Seidal Iterative method.

Use $x_1^{(0)} = x_2^{(0)} = 0$, $x_3^{(0)} = x_4^{(0)} = 1$ as the initial approximation. Do 8 iterations. Use comment statements to make the program understandable.

(4)

SECTION IV

4. Attempt any two parts :

- (a) Derive the interpolating polynomial for the function $f(x)$ whose tabular values are given by :

x	3.2	2.7	1.0	4.8
f(x)	22	17.8	14.2	38.3

Use it to calculate $f(3)$. (3½)

- (b) (i) Define central difference operator δ and averaging operator μ and prove that

$$\mu = \sqrt{1 + \frac{\delta^2}{4}}.$$

- (ii) Calculate the nth divided difference of

$$f(x) = \frac{1}{x}. \quad (3½)$$

- (c) Find the least square approximation of second degree for the discrete data

x	-2	-1	0	1	2
f(x)	15	1	1	3	19

(3½)

(1500)****